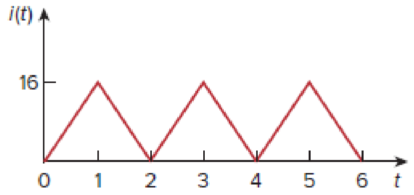
# Chapter 11, Problem 7PP

## Problem

Find the rms value of the current waveform of Fig. 11.15. If the current flows through a 9-Ω resistor, calculate the average power absorbed by the resistor.

**Figure 11.15:**




## Step-by-step solution

**Step 1** of 4

Consider the following current waveform:

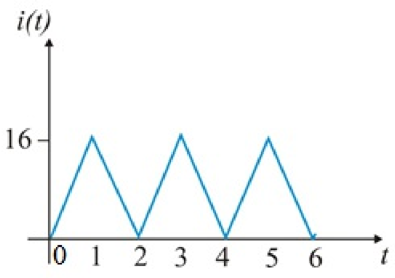



Figure 1

**Step 2** of 4

The time period of waveform from the waveform is $T=2$.

The expression for the current waveform in the period $0<t<1$ is,

$$i-0=\frac{16-0}{1-0}(t-0)$$
$$i=16t$$

The expression for the current waveform in the period $1<t<2$ is,

$$i-16=\frac{0-16}{2-1}(t-1)$$
$$\begin{aligned} i&=16-16(t-1)\\ &=16(2-t)\\ &=32-16t\\ &=16(2-t) \end{aligned}$$

Therefore, the expression of current waveform in one period is,

$$i(t)=\begin{cases}16t, & 0<t<1\\ 16(2-t), & 1<t<2\end{cases}$$

**Step 3** of 4

Calculate the rms value of current $I_{\text{rms}}$:

$$
\begin{aligned}
I_{\text{rms}} &= \sqrt{\frac{1}{T}\int_0^T i^2\,dt} \\
&= \sqrt{\frac{1}{2}\left(\int_0^1 (16t)^2\,dt + \int_1^2 16^2(2-t)^2\,dt\right)} \\
&= \sqrt{\frac{1}{2}\left(16^2\int_0^1 t^2 dt + 16^2\int_1^2 (2-t)^2\,dt\right)} \\
&= \sqrt{\frac{16^2}{2}\left(\int_0^1 t^2 dt + \int_1^2 \left(4+t^2-4t\right)dt\right)}
\end{aligned}
$$

$$
\begin{aligned}
&= \sqrt{\frac{16^2}{2}\left(\left.\frac{t^3}{3}\right|_0^1 + 4t\Big|_1^2 + \left.\frac{t^3}{3}\right|_1^2 - 4\left.\frac{t^2}{2}\right|_1^2\right)} \\
&= \sqrt{\frac{16^2}{2}\left(\frac{1^3-0^3}{3} + 4(2-1) + \frac{2^3-1^3}{3} - 4\frac{\left(2^2-1^2\right)}{2}\right)} \\
&= \sqrt{\frac{16^2}{2}\left(\frac{1}{3}+4+\frac{7}{3}-6\right)} \\
&= 9.238\text{ A}
\end{aligned}
$$

Therefore, the rms value of current $I_{\text{rms}}$ is $\boxed{9.238\text{ A}}$.

**Step 4** of 4

Calculate the average power absorbed by $9\ \Omega$ resistor.

$$
\begin{aligned}
P &= I_{\text{rms}}^2 R \\
&= (9.238\text{ A})^2 (9\ \Omega) \\
&= (9.238)^2 (9) \\
&= 768\text{ W}
\end{aligned}
$$

Therefore, the average power absorbed by $9\ \Omega$ resistor is $\boxed{768\text{ W}}$.